right now because as skeptical as we are if it works then standing in line in a supermarket would be talked about with an air of melancholy like our beloved cassettes.

Medical technologies

Medical technology has a myriad of slots which embedded systems can fit into. Take for example the pills that are equipped with 'smart' processors which assess the situation and accordingly repair organs or process information about cell formation or anomalies in cell operation. But that is in the near future – right now we have a system which has saved the lives of over two million people – the pacemaker-and it is all thanks to embedded systems.

A little history first.

In the 18th Century, it was already common information that electrical stimulation causes the muscles to contract and Charles Kite was the first to describe this idea, which is so familiar to us right now, of using electrical discharges to save people who have suffered a heart attack.

Based on this, the first pacemakers were built – but they were not exactly the ideal elixir to an ailing heart. They were large, bulky and ugly and were built around vacuum tubes and were powered by external power supply. Because of this design, it was impossible to be implanted inside a person.

The first external cardiac pacemaker

In 1947, the transistor was invented. This ensured a very bright possibility for implantable devices. In 1958, Arne Larsson received the first fully implantable pacemaker. Though the battery lasted for only three hours, it was a clear indicator that it was in-fact possible to artificially pace a human being's heart. The man did receive a total of 26 pacemakers, which made us wonder if he was just some experiment for the scientists, but he lived to a ripe age of 86.

So a normal heart has a pacemaker region where the pumping action is synchronised. This pacemaker produces electrical energy (yes the NATURAL pacemaker produces electrical energy) which creates an impulse. This impulse is transferred to the atria (a chamber in which blood enters the heart) which makes them contract and push the blood into the ventricles. After 150 milliseconds, the impulse then in turn moves to the ventricles which in turn